EPRC Methods Workshop: Difference-in-Differences

Draft Talk Outline and Data Example Plan

Draft working document

October 2026

Purpose

This document sketches the first draft of an EPRC methods workshop on difference-in-differences (DID) for education policy research. The immediate goal is to build Part 1 of the talk around a clean generated panel example, while motivating the method with concrete education policy questions. The running policy example should be a gifted and talented (G&T) program: do students who enter G&T later fare better than similar students who do not enter? That same example naturally sets up the staggered DID handoff, because students may enter G&T at different grade points.

Core Data Choice

The generated workshop data are intentionally simple and use the same basic structure as the simulation project: units observed over time, known treatment timing, and known treatment effects. The current files retain the district-year language from the simulation project, but the talk can describe the design using the G&T example. Conceptually, the unit could be a student, school, or district; the first pass should keep treatment assignment and analysis at the same level.

- Running example: G&T program entry.
- Unit in the generated data: district; unit in the motivating example: student.
- Time: 11 years
- Outcome: synthetic standardized achievement
- Treatment: program entry/exposure
- Simple DID timing: one treated cohort enters in year 5
- Staggered preview timing: early entrants in year 5 and later entrants in year 8

Generated Files

The current outline uses two district-level data sets.

Data set	Purpose
`simple_did_district`	Main Part 1 teaching data. Half of districts adopt in year 5 and half are never treated.
`staggered_did_district`	Light preview data for Wei's staggered DID section. Thirty percent enter treatment in year 5, twenty percent enter in year 8, and half are never treated.

Each data set is saved in both `.csv` and `.dta` formats.

Talk Arc

Part 1: Simple DID

The first part should emphasize intuition before notation. The key message is: DID compares the treated group's change over time to the comparison group's change over the same period.

1. Start with a policy question: do students who enter G&T later have better achievement outcomes?
2. Show the unit-year data structure.
3. Define treated units, comparison units, pre-periods, and post-periods.
4. Show the two-by-two DID contrast.
5. Connect the visual contrast to the regression coefficient.
6. Discuss the parallel trends assumption.
7. Use the known simulated effect to explain why generated data are useful pedagogically.

Part 2: Preview for Staggered DID

Wei's section can begin by relaxing the one-adoption-date setup. For now, this outline only needs a transition slide and placeholders. The treatment is still the same policy, but units enter treatment at different times.

- Early G&T entrants begin in year 5.
- Later G&T entrants begin in year 8.
- Never-treated/never-entering students remain available as one comparison group.
- The only figures needed at this stage are the timing plot and a simple group-trends plot showing both treated cohorts over time.

Draft Slide-by-Slide Outline

Slide	Main purpose	Figure or table to use
1	Motivate the workshop question with education policy examples.	Use a small examples table, with G&T entry as the running example.
2	Explain why DID is useful for policy timing.	Use a two-by-two G&T DID table: students who enter G&T versus comparison students, before and after entry.
3	Show the actual data structure.	Use a variable table mapping the G&T example to the generated data variables.
4	Show the simple one-cohort DID design.	Use Figure 1: mean trends for treated-in-year-5 and never-treated districts.
5	Teach the two-by-two DID contrast.	Use Figure 2: pre/post means for treated and comparison groups.
6	Connect the contrast to regression.	Use the DID regression equation and a tiny estimate-versus-truth table.
7	Discuss parallel trends.	Return to Figure 1 and focus attention on the pre-treatment years.
8	Hand off to staggered DID.	Use Figure 3: early adopters in year 5, later adopters in year 8, never treated.
9	Placeholder for Wei's section.	Use Figure 4: group trends for early-treated, later-treated, and never-treated units, plus placeholder bullets.

Slide 1: Workshop Question

Main point. DID is useful when we observe treated and comparison units before and after a policy change.

Visual/table. Use a short education-policy examples table. The G&T example should be the running example for the rest of the talk.

Policy question	Treatment	Outcome
Do students who enter G&T fare better?	Entry into gifted and talented programming	Achievement, course-taking, attendance
Did high-dosage tutoring improve learning?	School or student receives tutoring	Test scores or growth
Did a new curriculum improve math achievement?	District adopts the curriculum	District or student math outcomes
Did accountability pressure change outcomes?	School receives an accountability designation	Achievement, discipline, enrollment

Slide 2: What Data Do We Need?

Main point. A DID design needs units observed over time, treatment timing, and outcomes measured before and after treatment.

Visual/table. Use a small two-by-two design table for the simple G&T version of the example.

	Before G&T entry	**After G&T entry**
Students who enter G&T	Prior achievement trajectory	Later achievement after entry
Comparison students	Prior achievement trajectory	Later achievement without entry

Slide 3: The Generated Data Structure

Main point. Generated data let us know the truth while still looking like education panel data.

Visual/table. Use a variable table.

Variable	**G&T interpretation for the workshop**
`unit_id`	Student, school, or district identifier; the current generated file uses `district_id`.
`year`	Year or grade point in the panel.
`treatment_start_year`	First year/grade of G&T entry; 0 for never-entering units.
`cohort`	Entry-timing group, such as year 5 entrants, year 8 entrants, or never treated.
`treated`	Indicator for whether the unit is currently exposed to G&T.
`event_time`	Years since G&T entry.
`outcome`	Synthetic standardized achievement.
`true_effect`	Known effect added by the simulation.

Slide 4: The Simple DID Design

Main point. In Part 1, there is one treated cohort and one comparison group.

On slide.

- 50% of units enter treatment in year 5.
- 50% are never treated.
- Treatment stays on after adoption.
- We compare pre-to-post changes across groups.

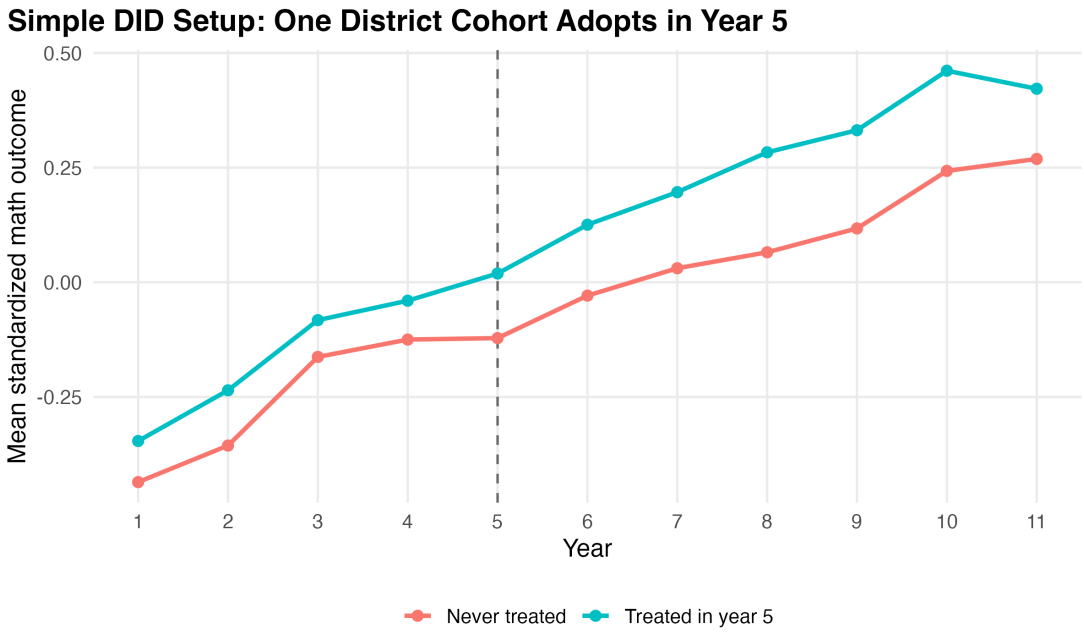


Figure 1: Figure 1. Simple DID design with one treated cohort.

Slide 5: Difference in Differences

Main point. DID is the difference between two changes.

On slide.

- Treated change: treated post mean minus treated pre mean.
- Comparison change: comparison post mean minus comparison pre mean.
- DID estimate: treated change minus comparison change.

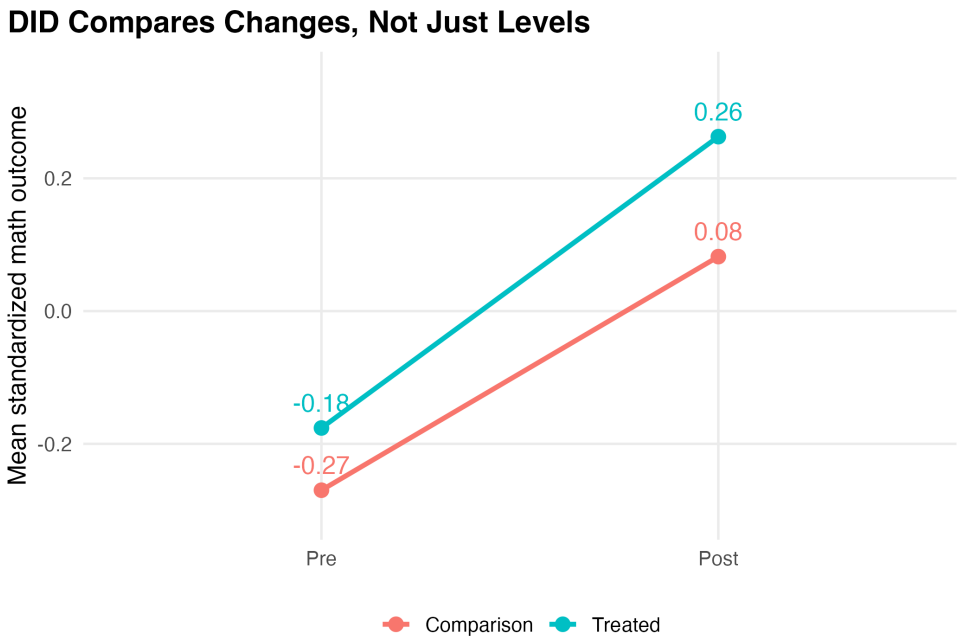


Figure 2: Figure 2. Two-by-two DID contrast.

Slide 6: Regression Version

Main point. The two-by-two DID contrast can be estimated with a simple interaction model.

$$Y_{it} = \alpha + \beta_1 \text{Treated}_i + \beta_2 \text{Post}_t + \delta(\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

On slide.

- δ is the DID estimate.
- The coefficient has the same logic as the visual contrast.
- In the generated data, the DID estimate is close to the known effect.

Model	Estimate	Known effect
Two-by-two DID regression	0.087	0.090
District and year fixed effects	0.087	0.090

Slide 7: The Identifying Assumption

Main point. DID requires a counterfactual claim: absent the treatment, treated districts would have followed the comparison group's trend.

On slide.

- Parallel trends is about untreated potential outcomes.
- We can inspect pre-trends, but we cannot directly prove the assumption.
- Use Figure 1 again and focus on years 1–4.

Slide 8: Transition to Staggered DID

Main point. Real education policies rarely start everywhere at the same time.

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- Same G&T policy/program.
- Some students enter in year 5.
- Other students enter in year 8.
- Some students never enter.

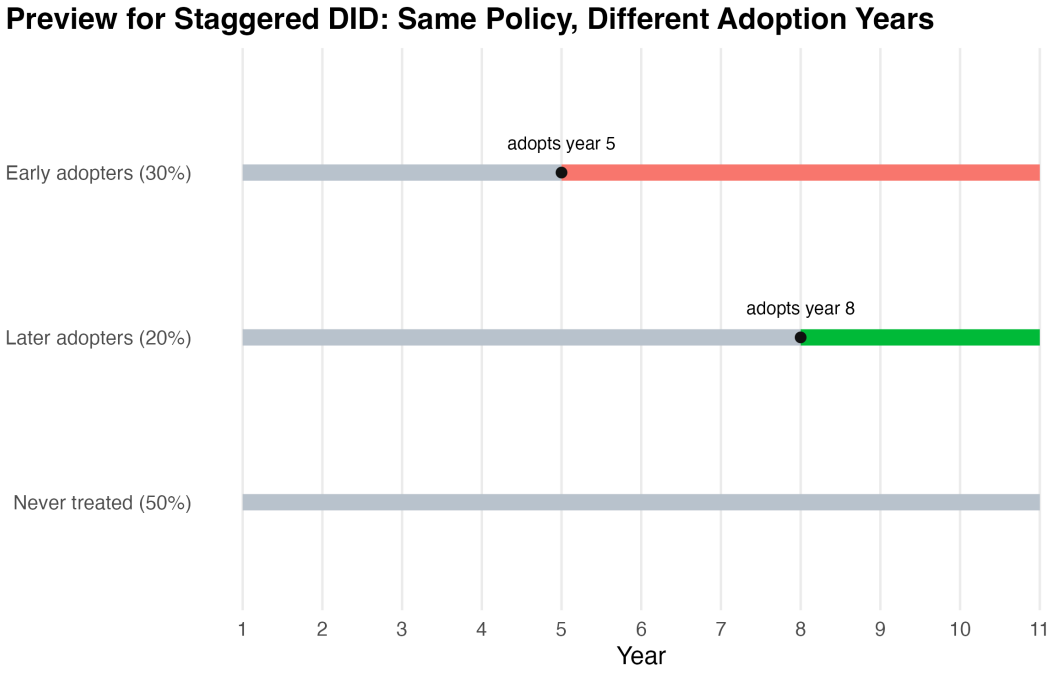


Figure 3: Figure 3. Treatment timing preview for the staggered DID section.

Slide 9: Placeholder for Wei's Staggered DID Section

Main point. Keep this light for now. This slide is a placeholder for Wei's fuller staggered DID section.

On slide.

- Use the same G&T question, but now students enter at different points.
- What changes when treatment timing varies?
- What is the comparison group for each entry cohort and time?
- Which estimand and estimator should Wei walk through?

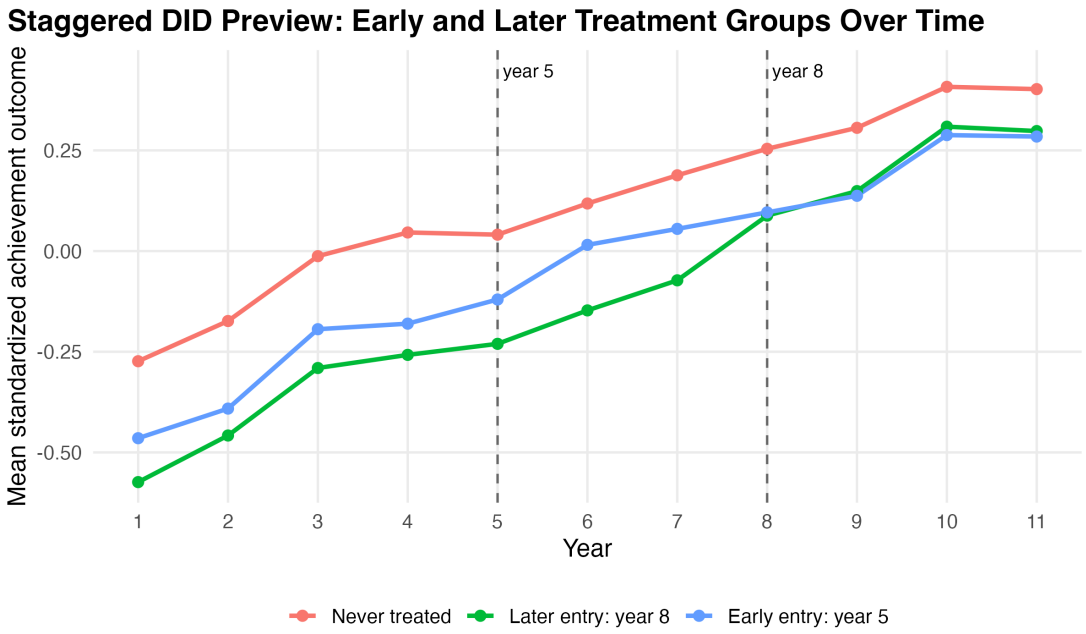


Figure 4: Figure 4. Group trends for the staggered DID preview.

Next Build Step

Once the workshop arc is settled, this outline can become a Beamer deck. The data and figures are already organized so the deck can pull from the same files.